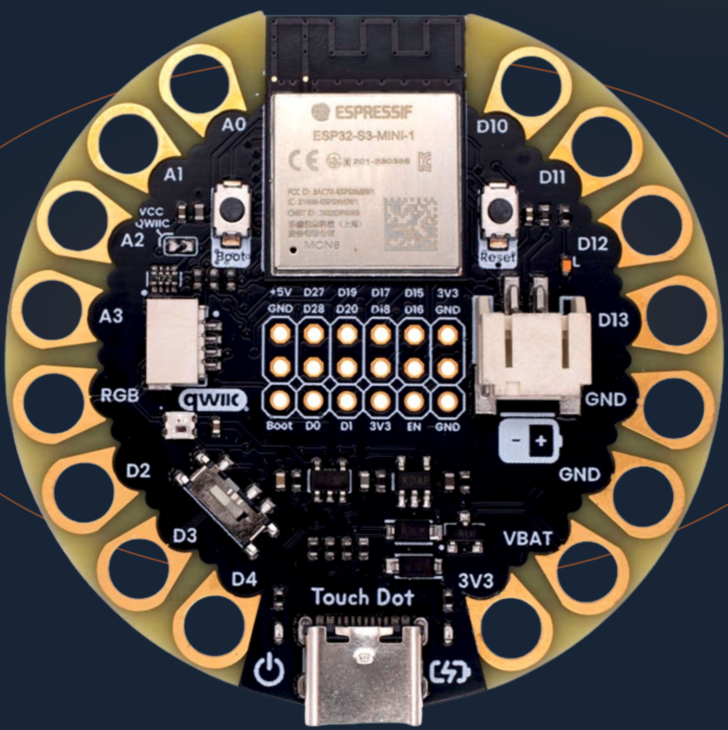
PRODUCT REFERENCE MANUAL

WEARABLE DEVELOPMENT BOARD

TouchDot S3

Mfr. Part #: UE0072

Version 1.0.0



ESP32-S3 wireless development, capacitive touch, battery power, storage, and sewable I/O in a compact circular board.

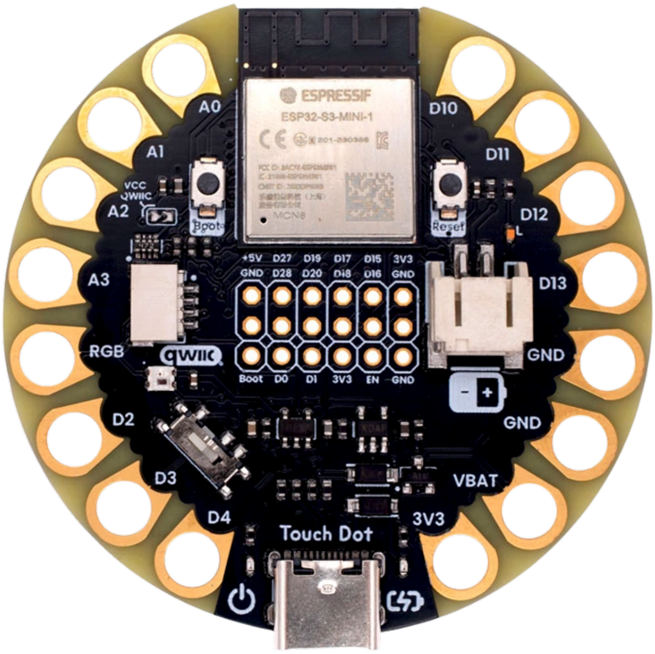
I²C / QWIIC | SPI / UART | USB-C / LiPo | Wi-Fi / BLE

DOCUMENT MAP

Contents

Description

The UNIT TouchDot S3 is a compact, circular development board based on the Espressif ESP32-S3-MINI-1 module. Its sewable-pad layout is intended for wearables, interactive textiles, IoT prototypes, and compact smart devices. The board combines 2.4 GHz Wi-Fi and Bluetooth Low Energy connectivity with capacitive-touch inputs, analog inputs, USB-C programming, battery operation, microSD storage, a QWIIC-compatible I²C connector, and onboard visual feedback.



Applications

- Wearable electronics and e-textiles
- Capacitive-touch user interfaces
- Battery-powered IoT nodes and data loggers
- Educational embedded-system projects
- Interactive art and compact connected devices

Software Support

TouchDot S3 can be programmed through the Arduino ecosystem, MicroPython, or Espressif ESP-IDF. The repository includes setup notes and examples for GPIO, ADC, I²C, SPI/microSD, the WS2812 LED, wireless communication, and deep sleep.

Hardware Features

- Espressif ESP32-S3-MINI-1-N8 module with dual-core Xtensa LX7 CPU
- 2.4 GHz 802.11 b/g/n Wi-Fi and Bluetooth Low Energy
- 16 sewable pads: 11 multiplexed GPIO pads, one NeoPixel data pad, two ground pads, one battery-voltage pad, and one 3.3 V pad
- Capacitive-touch and analog functions exposed on the sewable pads
- USB-C connector for power, native USB data, programming, and charging
- PH 2.0 mm Li-ion/LiPo battery connector and onboard charge management
- QWIIC-compatible 4-pin I^2C connector at 3.3 V
- microSD socket connected through SPI
- WS2812B-2020 addressable RGB LED and a user LED
- Boot and reset buttons, power switch, and power/charge indicators
- Expansion, serial-programming, JTAG, and USB test-point access

1 The Board

1.1 Package Contents

- 1 × TouchDot S3 development board (UE0072)

USB-C cable, Li-ion/LiPo battery, QWIIC cable, headers, microSD card, and external peripherals are not included unless stated by the distributor.

1.2 Board Identification

Item	Value
Product	UNIT TouchDot S3
Manufacturer part number	UE0072
Documented hardware revision	V0.1.2
Current released pinout artwork	V0.2.0
Main module	ESP32-S3-MINI-1-N8
Logic rail	3.3 V

The hardware-revision value identifies the released schematic and mechanical artwork. The pinout-artwork revision identifies the drawing itself and does not change the board hardware revision.

2 Electrical Characteristics

2.1 Documented Operating Values

Symbol	Description	Value	Unit
VUSB	USB-C input supply	5	V
VCC	Digital logic and QWIIC supply	3.3	V
VBAT	Single-cell Li-ion/LiPo battery input	Battery rail	—
ADCRES	ADC resolution used by the board documentation	12	bit
FI2C	Typical I^2C bus frequency	400	kHz

These are interface values documented by the project, not absolute maximum ratings. Refer to the ESP32-S3-MINI-1, regulator, charger, and peripheral component datasheets before designing near a component limit.

2.2 Electrical Precautions

- GPIO, touch, ADC, UART, SPI, and I^2C signals use 3.3 V logic. Do not apply 5 V directly to an ESP32-S3 GPIO.
- Keep analog input voltage within the 3.3 V supply domain.
- Use a protected, compatible single-cell Li-ion/LiPo battery with the correct PH 2.0 connector polarity shown in the pinout.
- Turn the board off before changing wiring or inserting/removing a microSD card.
- The 5 V expansion pin is a power rail, not a 5 V-tolerant logic reference.

2.3 Power Paths

The board can be powered from USB-C or the battery connector. The AP2112K LDO provides the 3.3 V rail. The power switch controls normal board operation, while the onboard charge circuit supports battery charging from USB-C. The red PWR and amber CHRG indicators show power and charging state respectively.

3 Functional Overview

3.1 Processing and Wireless Connectivity

The ESP32-S3-MINI-1-N8 module provides the application processor, flash storage, 2.4 GHz Wi-Fi, and Bluetooth Low Energy. The board exposes native USB D+ and D− signals through USB-C and dedicated test points.

3.2 Wearable I/O

Large plated holes around the board edge may be soldered, clipped, or sewn with conductive thread. The signal pads expose GPIO1–4, GPIO7–11, GPIO17, GPIO18, and GPIO45. GPIO1–4 and GPIO7–11 also provide analog and/or capacitive-touch functions as marked on the board pinout.

3.3 User Feedback and Controls

- **L3 / D13:** user LED on GPIO7
- **L4 / D25:** WS2812B-2020 RGB LED data input on GPIO45; its data output is routed to the D25 sewable pad for chaining compatible LEDs
- **PB1:** Boot button associated with GPIO0
- **PB2:** Reset button connected to EN
- **SW1:** board power switch
- **L1 / L2:** power and charge indicators

3.4 I²C / QWIIC Expansion

The four-pin QWIIC-compatible connector exposes GND, 3.3 V, SDA/GPIO5, and SCL/GPIO6. Solder bridge SB1 enables the connector supply. The signal order and 3.3 V domain must be checked before attaching a cable or third-party module.

3.5 microSD Storage

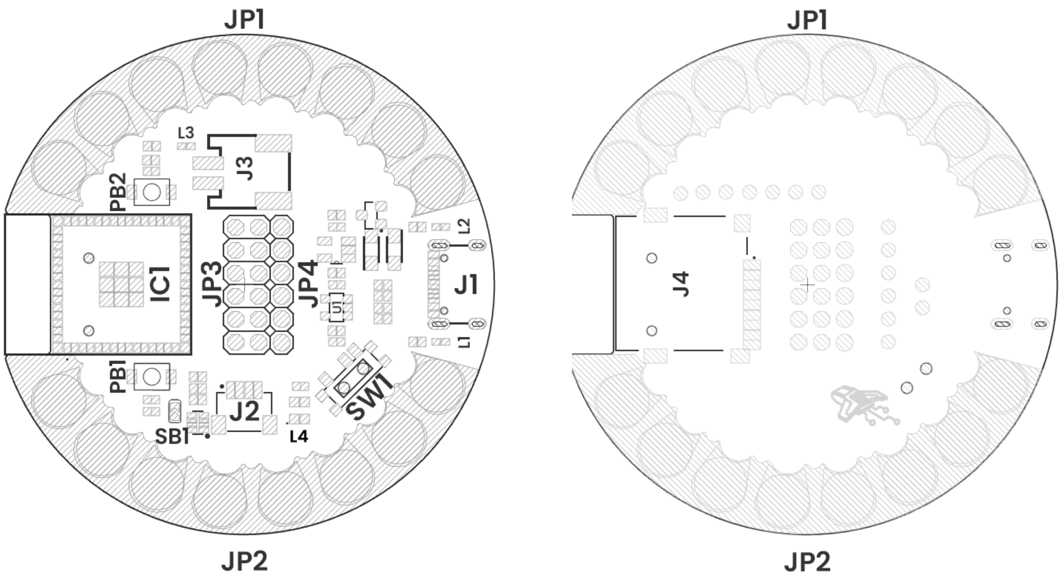
The bottom-side microSD socket operates in SPI mode. GPIO21 is chip select, GPIO9 is MOSI, GPIO7 is SCK, and GPIO8 is MISO. Pull-up resistors for GPIO8, GPIO9, and GPIO21 are associated with the microSD interface.

3.6 Battery and USB-C

The PH 2.0 connector accepts a compatible single-cell Li-ion/LiPo battery. USB-C supplies power, charges the battery through the onboard charge circuit, and connects the ESP32-S3 native USB interface for programming and data.

3.7 Debug and Expansion Access

The board includes a 2 × 6 expansion header, a six-pin serial-programming row, JTAG test points, and USB D+/D− test points. These interfaces expose additional GPIO and system signals while preserving the compact circular form factor.

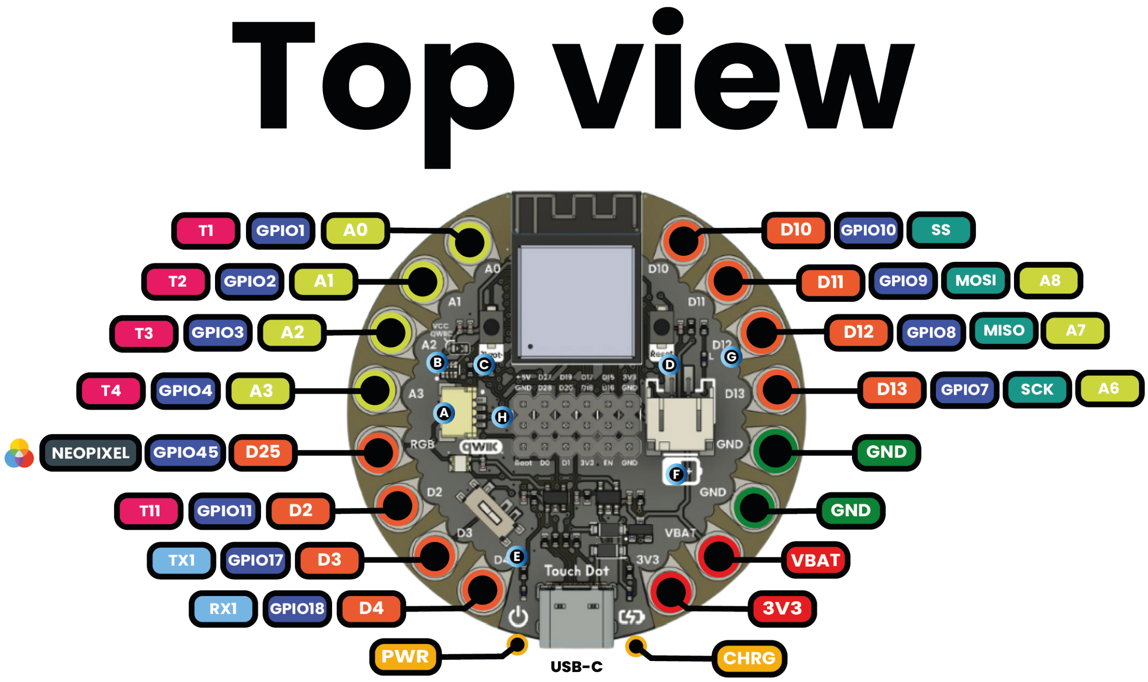


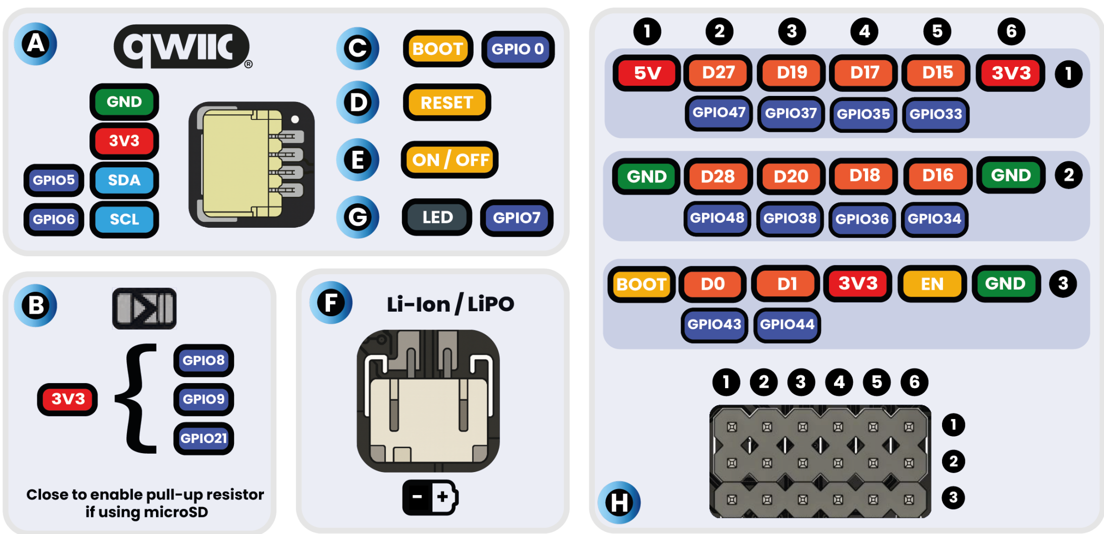
Views of Touch Dot S3 Topology

Reference	Description
IC1	Espressif ESP32-S3 module
U1	AP2112K 3.3 V LDO regulator
PB1 / PB2	Boot / reset push buttons
SW1	Power switch
L1 / L2	Power / charge indicators
L3	User LED on GPIO7 (D13)
L4	WS2812B-2020 RGB LED on GPIO45 (D25)
SB1	QWIIC VCC enable solder bridge
J1	USB-C connector
J2	QWIIC-compatible I^2C connector
J3	PH 2.0 battery connector
J4	microSD socket
JP1 / JP2	Sewable pads
JP3 / JP4	GPIO, system, and power headers

4 Connectors and Pinouts

4.1 Top and Bottom Pinout

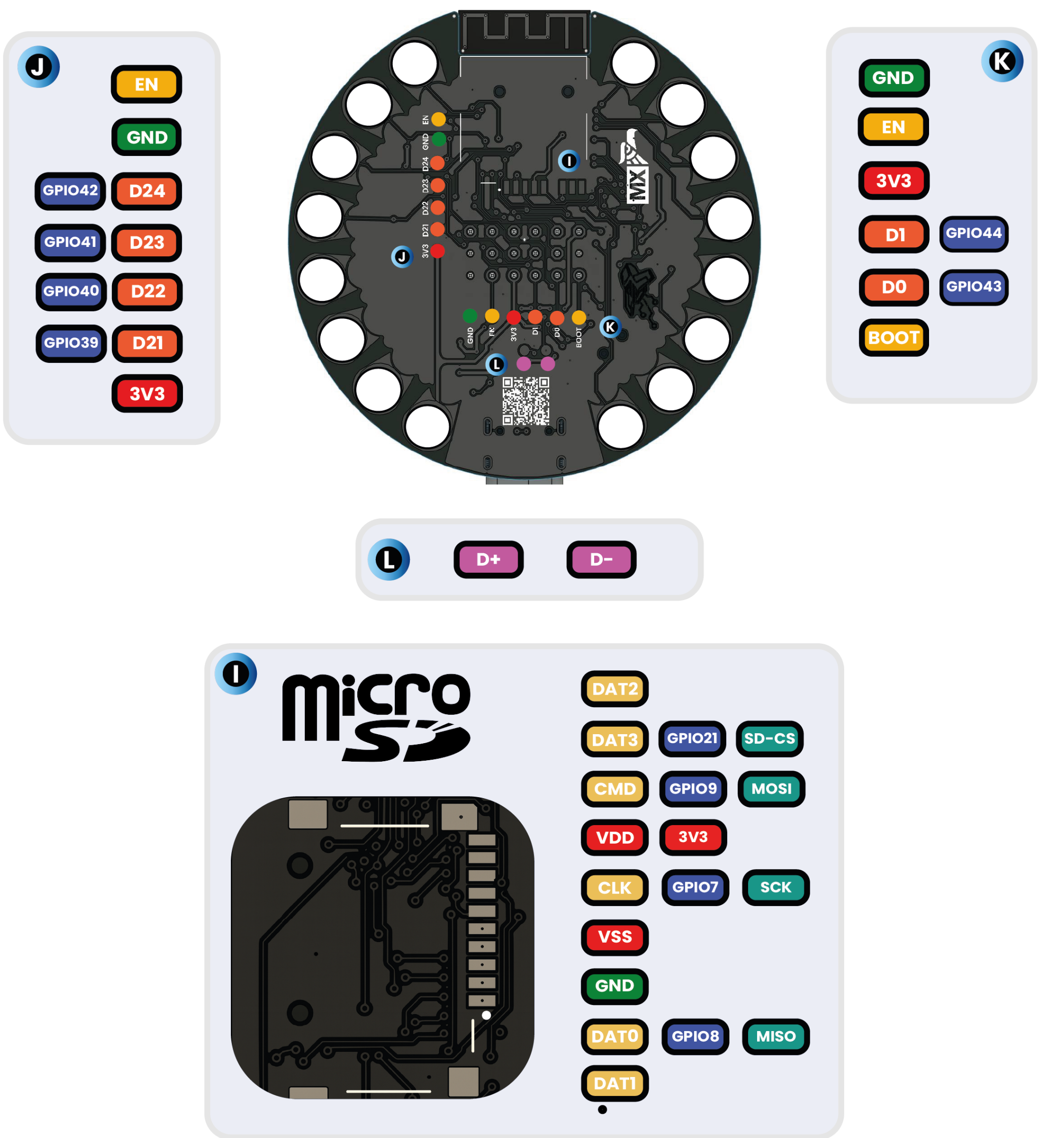




Description:

Supply voltage	GPIO compatible with Arduino	Touch pins	I2C
GND	GPIO ESP32	Analog pins	RGB
Components	UART	SPI	

Bottom view



Description:

- Supply voltage
- GND
- GPIO compatible with Arduino
- GPIO ESP32
- Micro SD Data
- SPI
- USB

4.2 Sewable Pads

Board label	ESP32-S3 signal	Additional function
A0 / T1	GPIO1	ADC, touch
A1 / T2	GPIO2	ADC, touch

Board label	ESP32-S3 signal	Additional function
A2 / T3	GPIO3	ADC, touch
A3 / T4	GPIO4	ADC, touch
D25 / NeoPixel	GPIO45	WS2812 data output
D2 / T11	GPIO11	Touch
D3 / TX1	GPIO17	UART1 TX
D4 / RX1	GPIO18	UART1 RX
D10 / SS / T10	GPIO10	SPI chip select, touch
D11 / MOSI / A8 / T9	GPIO9	SPI MOSI, ADC, touch
D12 / MISO / A7 / T8	GPIO8	SPI MISO, ADC, touch
D13 / SCK / A6 / T7	GPIO7	SPI clock, ADC, touch, user LED
GND (2 pads)	GND	Ground return
VBAT	Battery rail	Battery supply access
3V3	3.3 V rail	Regulated supply access

4.3 QWIIC-Compatible Connector (J2)

Pin	Signal	ESP32-S3 GPIO
1	GND	—
2	3.3 V	—
3	SDA / A4 / T5	GPIO5
4	SCL / A5 / T6	GPIO6

4.4 microSD Socket (J4, SPI Mode)

microSD pin	Name	SPI function	ESP32-S3 GPIO
1	DAT2	Not used in SPI mode	—
2	DAT3 / CD	CS	GPIO21
3	CMD	MOSI	GPIO9
4	VDD	3.3 V	—
5	CLK	SCK	GPIO7
6	VSS	GND	—
7	DAT0	MISO	GPIO8
8	DAT1	Not used in SPI mode	—

4.5 Expansion Header (2 × 6)

Position	Row 1	Row 2
1	5 V	GND
2	D27 / GPIO47	D28 / GPIO48
3	D19 / GPIO37	D20 / GPIO38
4	D17 / GPIO35	D18 / GPIO36
5	D15 / GPIO33	D16 / GPIO34
6	3.3 V	GND

4.6 Serial Programming Header (1 × 6)

Pin	Signal	ESP32-S3 connection
1	GND	GND
2	EN / Reset	EN
3	3.3 V	3.3 V rail
4	D1	GPIO44 (RX0)
5	D0	GPIO43 (TX0)
6	Boot	GPIO0

The D0/D1 labels follow the released board artwork. When connecting an external serial adapter, cross its TX and RX lines according to signal direction and verify the 3.3 V logic level.

4.7 JTAG and USB Test Points

Label	Signal
D21	GPIO39 / MTCK
D22	GPIO40 / MTDO
D23	GPIO41 / MTDI
D24	GPIO42 / MTMS
D+	USB D+
D−	USB D−

5 Board Operation

5.1 First Power-Up

1. Inspect the board for damage, debris, or shorts.
2. With the power switch off, connect a data-capable USB-C cable to a computer or regulated 5 V USB supply.
3. Move the power switch to ON. The PWR indicator should illuminate.
4. Select an ESP32-S3 target in the chosen development environment and select the board's USB serial port.
5. If automatic download mode does not start, hold **Boot**, momentarily press **Reset**, and then release **Boot**.

5.2 Arduino LED Test

The following sketch checks the user LED on D13/GPIO7. GPIO7 is also the microSD clock, so do not run this test while accessing a card.

```
constexpr int LED_PIN = 7;

void setup() {
  pinMode(LED_PIN, OUTPUT);
}

void loop() {
  digitalWrite(LED_PIN, HIGH);
  delay(500);
  digitalWrite(LED_PIN, LOW);
  delay(500);
}
```

5.3 QWIIC / I²C Test

The default documented I²C pins are SDA on GPIO5 and SCL on GPIO6. Confirm that SB1 enables QWIIC VCC before expecting the connector to power a peripheral.

```
#include <Wire.h>

void setup() {
  Serial.begin(115200);
  Wire.begin(5, 6);

  for (uint8_t address = 1; address < 127; ++address) {
    Wire.beginTransmission(address);
    if (Wire.endTransmission() == 0) {
      Serial.printf("I2C device: 0x%02X\n", address);
    }
  }
}

void loop() {}
```

5.4 microSD Use

Use SPI pins CS=GPIO21, MOSI=GPIO9, MISO=GPIO8, and SCK=GPIO7. Format a supported card with FAT32 before use. Because GPIO7 also drives the user LED and GPIO7–9 are exposed on sewable pads, attached circuits must not contend with the microSD bus.

5.5 Battery Operation

Turn the board off before attaching a battery. Check connector polarity against the released pinout, insert a compatible single-cell Li-ion/LiPo battery, and then turn the power switch on. When USB-C is connected, the CHRG indicator may illuminate while the onboard circuit charges the battery. Do not use a damaged, swollen, reversed-polarity, or incompatible battery.

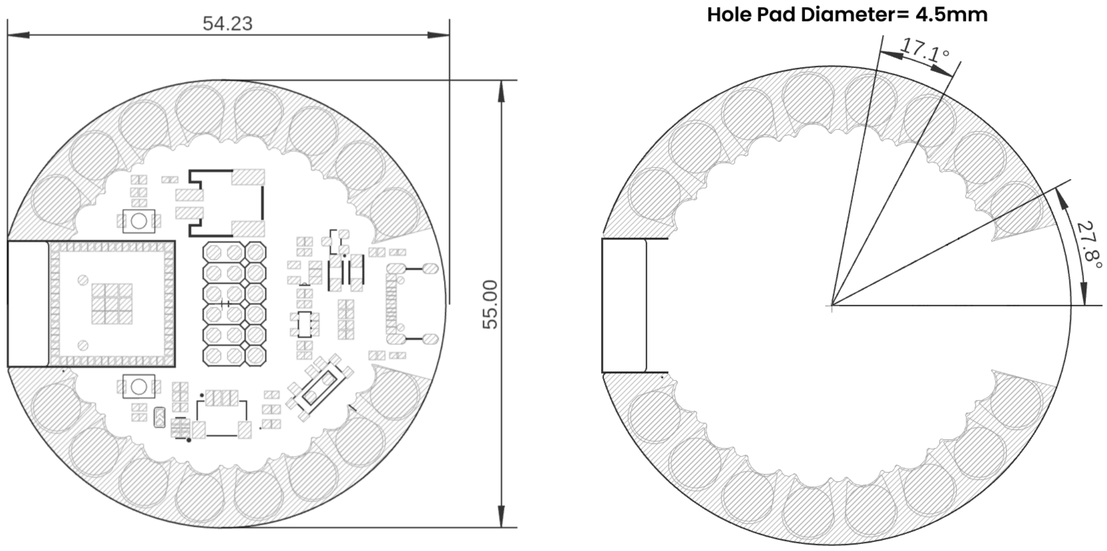
5.6 Shared-Pin Considerations

The ESP32-S3 pin matrix permits multiple functions on a GPIO, but a physical pin can only serve compatible roles at a given time. In particular:

- GPIO7 is D13, A6/T7, user LED, and microSD SCK.
- GPIO8 is D12, A7/T8, and microSD MISO.
- GPIO9 is D11, A8/T9, and microSD MOSI.
- GPIO21 is the microSD chip-select signal.
- GPIO5 and GPIO6 are the default QWIIC I^2C bus.
- GPIO45 drives the onboard WS2812 and the NeoPixel chain-output pad.

Plan firmware and external wiring so these functions do not conflict.

6 Mechanical Information



Mechanical dimensions in millimeters of Touch Dot S3

All dimensions are in millimeters unless otherwise noted. Use the released dimension drawing as the controlling reference for enclosure, textile, and mounting design. Allow clearance for the USB-C plug, battery connector, microSD card insertion, power switch, and any components soldered to the sewable pads or expansion headers.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	https://uelectronics.com/
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

Product specifications and documentation may be revised. Use the repository and product page listed in the next chapter for the latest released material.

8 Reference Documentation

Reference	Link
Product wiki	wiki.uelectronics.com/wiki/unit-touchdot-s3
GitHub repository	github.com/UNIT-Electronics-MX/unit_touchdot_s3
Product page	uelectronics.com/producto/unit-touchdot-s3
Hardware resources	repository hardware directory
ESP32-S3 documentation	Espressif ESP32-S3
Arduino IDE	arduino.cc/en/software
Thonny IDE	thonny.org
Visual Studio Code	code.visualstudio.com

9 Appendix

9.1 Schematic

The released V0.1.2 schematic is available in the repository:

TouchDot S3 schematic PDF

9.2 Source Hierarchy and Revision Notes

This product reference consolidates the maintained repository README, hardware README, V0.1.2 schematic and mechanical/topology artwork, V0.2.0 pinout artwork, and the existing programming documentation. If values conflict, use the newest released hardware source for the applicable board revision and verify the signal on the physical board before connecting external hardware.

The serial D0/D1 Arduino labels differ between some legacy text and the current V0.2.0 pinout artwork. This document follows the V0.2.0 artwork for its header table and retains the underlying GPIO numbers so users can identify the signal unambiguously.

9.3 Document Control

Field	Value
Product	TouchDot S3
Manufacturer part number	UE0072
Hardware revision covered	V0.1.2
Pinout artwork used	V0.2.0
Product-reference revision	1.0.0
Publication date	2026-07-31